1 Variational Inequality

Inside the supremum:

$$P_t^X + S_t P_t^Y + R_t = \int_0^t \left(\varphi_c(Y_{s^-} - \xi) - \varphi_c(Y_{s^-}) - \xi S_t + r(Y_{s^-})\right) dN_s^a + \int_0^t \left(\varphi_c(Y_{s^-} + \xi) - \varphi_c(Y_{s^-}) + \xi S_t + r(Y_{s^-})\right) dN_s^b$$

Value function, at time t for a stopping time τ in $[t, T]$:

$$v(t, y, S) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\left[\int_t^\tau \beta_\tau^a(Y_{s^-}, S_s)\, dN_s^a + \int_t^\tau \beta_\tau^b(Y_{s^-}, S_s)\, dN_s^b\right]$$

where:

$$\beta_\tau^a(Y_{s^-}, S_s) = \varphi_c(Y_{s^-} - \xi) - \varphi_c(Y_{s^-}) - \xi S_\tau + \mathfrak{r}(Y_{s^-})$$
$$\beta_\tau^b(Y_{s^-}, S_s) = \varphi_c(Y_{s^-} + \xi) - \varphi_c(Y_{s^-}) + \xi S_\tau + \mathfrak{r}(Y_{s^-})$$

Dynamic principle, we stop at time t ($\tau = t$) or we continue ($\tau > t$):

$$v(t, y, S) = \max\left\{\mathbb{E}\left[\int_t^{t+h} \beta_{t+h}^b(Y_{s^-}, S_s)\, dN_s^b + \int_t^{t+h} \beta_{t+h}^a(Y_{s^-}, S_s)\, dN_s^a + v(t+h, Y_\tau^{t, Y_{t+h}^{t,y}}, S_\tau^{t, S_{t+h}^{t,S}}) \;\middle|\; \mathcal{F}_t\right], 0\right\}$$
$$0 = \max\left\{\mathbb{E}\left[\int_t^{t+h} \beta_{t+h}^b(Y_{s^-}, S_s)\, dN_s^b + \int_t^{t+h} \beta_{t+h}^a(Y_{s^-}, S_s)\, dN_s^a + v(t+h, Y_\tau^{t, Y_{t+h}^{t,y}}, Z_\tau^{t, Z_{t+h}^{t,z}}, S_\tau^{t, S_{t+h}^{t,S}}) - v(t, Y_\tau^{t,y}, S_\tau^{t,s}) \;\middle|\; \mathcal{F}_t\right], -v(t, y, S)\right\}$$
$$0 = \max\left\{\mathbb{E}\left[\int_t^{t+h} \beta_{t+h}^b(Y_{s^-}, S_s)\, dN_s^b + \int_t^{t+h} \beta_{t+h}^a(Y_{s^-}, S_s)\, dN_s^a + \int_t^{t+h} dv(u, Y_u, S_u) \;\middle|\; \mathcal{F}_t\right], -v(t, y, S)\right\}$$

Applying Itô formula on $v(t, Y_t, S_t)$ with jump and diffusion processes:

$$\begin{aligned}
0 = \max\Big\{\mathbb{E}\Big[&\int_t^{t+h} \frac{\partial}{\partial t} v(u, Y_u, S_u)\, du + \int_t^{t+h} \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(u, Y_u, S_u)\, du + \int_t^{t+h} \sigma \frac{\partial}{\partial S} v(u, Y_u, S_u)\, dW_u \\
&+ \int_t^{t+h} \left(\beta_{t+h}^a(Y_{u^-}, S_u) + v(u, Y_{u^-} - \xi, S_u) - v(u, Y_{u^-}, S_u)\right) dN_u^a \\
&+ \int_t^{t+h} \left(\beta_{t+h}^b(Y_{u^-}, S_u) + v(u, Y_{u^-} + \xi, S_u) - v(u, Y_{u^-}, S_u)\right) dN_u^b \;\Big|\; \mathcal{F}_t\Big], -v(t, y, S)\Big\}
\end{aligned}$$

Recalling that $d\tilde{N}_t^a = dN_t^a - \lambda_t^a\, dt$, $d\tilde{N}_t^b = dN_t^b - \lambda_t^a\, dt$ with $\lambda_t^a = \bar{\lambda}^a(Y_{t^-}, S_t)$, $\bar{\lambda}^b(Y_{t^-}, S_t)$.

Expectation of martingale parts $\int .dW_t$ and $\int .d\tilde{N}_t$ are 0.

$$\begin{aligned}
0 = \max\Big\{&\int_t^{t+h} \frac{\partial}{\partial t} v(u, y, S) + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(u, y, S) \\
&+ \bar{\lambda}^a(y, S)\big[\varphi_c(y - \xi) - \varphi_c(y) - \xi S_{t+h} + r(y) + v(u, y - \xi, S) - v(u, y, S)\big] \\
&+ \bar{\lambda}^b(y, S)\big[\varphi_c(y + \xi) - \varphi_c(y) + \xi S_{t+h} + r(y) + v(u, y + \xi, S) - v(u, y, S)\big]\, du, -v(t, y, S)\Big\}
\end{aligned}$$

Therefore, for h small enough, the variational inequality is:

$$\begin{aligned}
0 = \max\Big\{&\frac{\partial}{\partial t} v(t, y, S) + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(t, y, S) \\
&+ \bar{\lambda}^a(y, S)\big[\varphi_c(y - \xi) - \varphi_c(y) - \xi S_t + r(y) + v(t, y - \xi, S) - v(t, y, S)\big] \\
&+ \bar{\lambda}^b(y, S)\big[\varphi_c(y + \xi) - \varphi_c(y) + \xi S_t + r(y) + v(t, y + \xi, S) - v(t, y, S)\big], -v(t, y, S)\Big\}
\end{aligned}$$

Tentative with risk-averse framework:
With terminal condition : $v(T, y, S) = -\exp\left\{-\gamma\left(x - X_0 + (y - Y_0)S + r(y)\right)\right\}$

2 Euler Scheme - Risk Neutral

We have x, y variables linked, $\frac{x}{y} = z$ and $\varphi_c(y) = \frac{c}{y} = x$. Therefore, the problem can be reduce from dimension 4 to dimension 2, we write again the EDP in the variational inequality.

$$\begin{aligned}
0 =& \frac{\partial}{\partial t} v(t,y,S) + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(t,y,S) \\
&+ \bar{\lambda}^a(y,S)\big[\varphi_c\big(y-\xi\big) - \varphi_c\big(y\big) - \xi S_t + r\big(\frac{c}{y^2}\big) + v(t,y-\xi,S) - v(t,y,S)\big] \\
&+ \bar{\lambda}^b(y,S)\big[\varphi_c\big(y+\xi\big) - \varphi_c\big(y\big) + \xi S_t + r\big(\frac{c}{y^2}\big) + v(t,y+\xi,S) - v(t,y,S)\big]
\end{aligned}$$

Backward Euler discretization for δ and h small enough and ξ for the jumps:

Diffusion part for $i \in \{0, ..., I-1\}$, $t_0 = 0$ and $t_I = T$ and $j \in \{1, ..., J-1\}$:

$$\frac{\partial}{\partial t} v(t,y,S) \approx \frac{v(t+\delta,y,S) - v(t,y,S)}{\delta} = \frac{v(t_{i+1},y_l,S_j) - v(t_i,y_l,S_j)}{\delta}$$
$$\frac{\partial^2}{\partial S^2} v(t,y,S) \approx \frac{v(t,y,S+h) - 2v(t,y,S) + v(t,y,S-h)}{h^2} = \frac{v(t_i,y_l,S_{j+1}) - 2v(t_i,y_l,S_j) + v(t_i,y_l,S_{j-1})}{h^2}$$
$$\frac{\partial}{\partial S} v(t,y,S) \approx \frac{v(t,y,S+h) - v(t,y,S)}{h}$$

Jump part for $l \in \{1, ..., L-1\}$:

$$\bar{\lambda}^a(y_l,S_j)\big[\varphi_c\big(y_{l-1}\big) - \varphi_c\big(y_l\big) - \xi S_j + r\big(\frac{c}{y_l^2}\big) + v(t_i,y_{l-1},S_j) - v(t_i,y_l,S_j)\big]$$
$$\bar{\lambda}^b(y_l,S_j)\big[\varphi_c\big(y_{l+1}\big) - \varphi_c\big(y_l\big) + \xi S_j + r\big(\frac{c}{y_l^2}\big) + v(t_i,y_{l+1},S_j) - v(t_i,y_l,S_j)\big]$$

<u>Full Euler scheme:</u>

$$\begin{aligned}
v(t_{i+1},y_l,S_j) = v(t_i,y_l,S_j)&\big[1 + \delta\big(\frac{\sigma^2}{h^2} + \bar{\lambda}^a(y_l,S_j) + \bar{\lambda}^b(y_l,S_j)\big)\big] \\
&+ v(t_i,y_l,S_{j+1})\big[-\frac{\delta\sigma^2}{2h^2}\big] \\
&+ v(t_i,y_l,S_{j-1})\big[-\frac{\delta\sigma^2}{2h^2}\big] \\
&+ v(t_{i+1},y_{l-1},S_j)\big[-\delta\bar{\lambda}^a(y_l,S_j)\big] \\
&+ v(t_{i+1},y_{l+1},S_j)\big[-\delta\bar{\lambda}^b(y_l,S_j)\big] \\
&- \delta\bar{\lambda}^a(y_l,S_j)\big[\varphi_c\big(y_{l-1}\big) - \varphi_c\big(y_l\big) - \xi S_j + r\big(\frac{c}{y_l^2}\big)\big] - \delta\bar{\lambda}^b(y_l,S_j)\big[\varphi_c\big(y_{l+1}\big) - \varphi_c\big(y_l\big) + \xi S_j + r\big(\frac{c}{y_l^2}\big)\big]
\end{aligned}$$

Operator Splitting Idea: At each time of the discretization, add jumps and then solve the linear system.

<u>Boundary conditions:</u>

A terminal time T, we have $v(T,y,S) = 0$, in the Euler scheme this means:

$\forall\, l \in \{0, \ldots, L\},\ j \in \{0, \ldots, J\}, v(t_I, y_l, S_j) = 0$

Neumann condition: $\frac{\partial^2 v(t,y,S)}{\partial S^2}\Big|_{S=S_0} = 0, \quad \frac{\partial^2 v(t,y,S)}{\partial S^2}\Big|_{S=S_J} = 0$, in the Euler scheme these mean:

$$
\begin{aligned}
v(t_{i+1}, y_l, S_0) = {} & v(t_i, y_l, S_0)\big[1 + \delta\big(\bar{\lambda}^a(y_l, S_0) + \bar{\lambda}^b(y_l, S_0)\big)\big] \\
& + v(t_i, y_{l-1}, S_0)\big[-\bar{\lambda}^a(y_l, S_0)\big] \\
& + v(t_i, y_{l+1}, S_0)\big[-\bar{\lambda}^b(y_l, S_0)\big] \\
& - \bar{\lambda}^a(y_l, S_0)\big[\varphi_c\big(y_{l-1}\big) - \varphi_c\big(y_l\big) - \xi S_0 + r\big(\frac{c}{y_l^2}\big)\big] - \bar{\lambda}^b(y_l, S_0)\big[\varphi_c\big(y_{l+1}\big) - \varphi_c\big(y_l\big) + \xi S_0 + r\big(\frac{c}{y_l^2}\big)\big] \\
v(t_{i+1}, y_l, S_J) = {} & v(t_i, y_l, S_J)\big[1 - \delta\big(\bar{\lambda}^a(y_l, S_J) + \bar{\lambda}^b(y_l, S_J)\big)\big] \\
& + v(t_i, y_{l-1}, S_J)\big[-\bar{\lambda}^a(y_l, S_J)\big] \\
& + v(t_i, y_{l+1}, S_J)\big[-\bar{\lambda}^b(y_l, S_J)\big] \\
& - \bar{\lambda}^a(y_l, S_J)\big[\varphi_c\big(y_{l-1}\big) - \varphi_c\big(y_l\big) - \xi S_J + r\big(\frac{c}{y_l^2}\big)\big] - \bar{\lambda}^b(y_l, S_J)\big[\varphi_c\big(y_{l+1}\big) - \varphi_c\big(y_l\big) + \xi S_J + r\big(\frac{c}{y_l^2}\big)\big]
\end{aligned}
$$

We denote:

$$
\begin{aligned}
c_0(y_l, S_j) &= \big[1 + \delta\big(\bar{\lambda}^a(y_l, S_j) + \bar{\lambda}^b(y_l, S_j)\big)\big] \\
c_1(y_l, S_j) &= -\frac{\delta\sigma^2}{2h^2} = c_1 \\
c_2(y_l, S_j) &= \big[1 + \delta\big(\frac{\sigma^2}{h^2} + \bar{\lambda}^a(y_l, S_j) + \bar{\lambda}^b(y_l, S_j)\big)\big] \\
c_3(y_l, S_j) &= -\frac{\delta\sigma^2}{2h^2} = c_3
\end{aligned}
$$

Linear system to solve: $V^{i+1,l} = AV^{i,l}$ for $i \in \{0, ..., I-1\}$ where:

$$
A = \begin{bmatrix}
c_0(y_l, S_0) & 0 & 0 & \cdots & & 0 \\
c_1 & c_2(y_l, S_1) & c_3 & \ddots & & \vdots \\
0 & c_1 & c_2(y_l, S_2) & \ddots & & \\
\vdots & \ddots & \ddots & \ddots & c_3 & 0 \\
 & & & c_1 & c_2(y_l, S_{J-1}) & c_3 \\
0 & \cdots & & 0 & 0 & c_0(y_l, S_J)
\end{bmatrix} \in \mathbb{R}^{J+1,J+1}, \qquad
V^{i,l} = \begin{bmatrix}
v(t_i, y_l, S_0) \\ v(t_i, y_l, S_1) \\ \\ \vdots \\ \\ v(t_i, y_l, S_{J-1}) \\ v(t_i, y_l, S_J)
\end{bmatrix} \in \mathbb{R}^{J+1}
$$

3 Euler Scheme - Risk Averse

We have x, y variables linked, $\frac{x}{y} = z$ and $\varphi_c(y) = \frac{c}{y} = x$. Therefore, the problem can be reduce from dimension 4 to dimension 2, we write again the EDP in the variational inequality.

$$
\begin{aligned}
0 = {} & \frac{\partial}{\partial t} v(t, y, S) + \frac{1}{2}\sigma^2 \left(\frac{\partial^2}{\partial S^2} v(t, y, S) - \psi \left(\frac{\partial v(t, y, S)}{\partial S} \right)^2 \right) \\
& + \mathbb{1}_{\{y+\xi \leq \overline{Y}\}} \bar{\lambda}^b(y, S) \frac{1}{\psi} \left[1 - \exp\left(-\psi \left(\varphi_c\big(y + \xi\big) - \varphi_c\big(y\big) + \xi S_t + r\big(y\big) + v(t, y + \xi, S) - v(t, y, S) \right) \right) \right] \\
& + \mathbb{1}_{\{y-\xi \geq \underline{Y}\}} \bar{\lambda}^a(y, S) \frac{1}{\psi} \left[1 - \exp\left(-\psi \left(\varphi_c\big(y - \xi\big) - \varphi_c\big(y\big) - \xi S_t + r\big(y\big) + v(t, y - \xi, S) - v(t, y, S) \right) \right) \right]
\end{aligned}
$$

Backward Euler discretization for δ and h small enough and ξ for the jumps:

Diffusion part for $i \in \{0, ..., I-1\}$, $t_0 = 0$ and $t_I = T$ and $j \in \{1, ..., J-1\}$:

$$\frac{\partial}{\partial t}v(t,y,S) \approx \frac{v(t+\delta,y,S)-v(t,y,S)}{\delta} = \frac{v(t_{i+1},y_l,S_j)-v(t_i,y_l,S_j)}{\delta}$$
$$\frac{\partial^2}{\partial S^2}v(t,y,S) \approx \frac{v(t,y,S+h)-2v(t,y,S)+v(t,y,S-h)}{h^2} = \frac{v(t_i,y_l,S_{j+1})-2v(t_i,y_l,S_j)+v(t_i,y_l,S_{j-1})}{h^2}$$
$$\frac{\partial}{\partial S}v(t,y,S) \approx \frac{v(t,y,S+h)-v(t,y,S-h)}{h} = \frac{v(t_i,y_l,S_{j+1})-v(t_i,y_l,S_{j-1})}{2h}$$
$$\left(\frac{\partial}{\partial S}v(t,y,S)\right)^2 \approx \left(\frac{v(t,y,S+h)-v(t,y,S-h)}{2h}\right)^2 = \frac{v^2(t_i,y_l,S_{j+1})-2v(t_i,y_l,S_{j+1})v(t_i,y_l,S_{j-1})+v^2(t_i,y_l,S_{j-1})}{4h^2}$$

Jump part for $l \in \{1,...,L-1\}$:

$$\bar{\lambda}^a(y_l,S_j)\frac{1}{\psi}\left[1-\exp\left(-\psi\left(\varphi_c(y_{l-1})-\varphi_c(y_l)-\xi S_j+r(y)+v(t_{i+1},y_{l-1},S_j)-v(t_{i+1},y_l,S_j)\right)\right)\right]$$
$$\bar{\lambda}^b(y_l,S_j)\frac{1}{\psi}\left[1-\exp\left(-\psi\left(\varphi_c(y_{l+1})-\varphi_c(y_l)+\xi S_j+r(y)+v(t_{i+1},y_{l+1},S_j)-v(t_{i+1},y_l,S_j)\right)\right)\right]$$

<u>Full Euler scheme:</u>

$$\begin{aligned}
v(t_{i+1},y_l,S_j) = {} & v(t_i,y_l,S_j)\left[1+\delta\frac{\sigma^2}{h^2}\right] \\
& + v(t_i,y_l,S_{j+1})\left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}\left(v(t_i,y_l,S_{j+1})-v(t_i,y_l,S_{j-1})\right)\right] \\
& + v(t_i,y_l,S_{j-1})\left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}\left(-v(t_i,y_l,S_{j+1})+v(t_i,y_l,S_{j-1})\right)\right] \\
& + \delta\bar{\lambda}^a(y_l,S_j)\frac{1}{\psi}\left[1-\exp\left(-\psi\left(\varphi_c(y_{l-1})-\varphi_c(y_l)-\xi S_j+r(y)+v(t_{i+1},y_{l-1},S_j)-v(t_{i+1},y_l,S_j)\right)\right)\right] \\
& + \delta\bar{\lambda}^b(y_l,S_j)\frac{1}{\psi}\left[1-\exp\left(-\psi\left(\varphi_c(y_{l+1})-\varphi_c(y_l)+\xi S_j+r(y)+v(t_{i+1},y_{l+1},S_j)-v(t_{i+1},y_l,S_j)\right)\right)\right]
\end{aligned}$$

Let us define the following function:

$$F(v_{i,l}) = v_{i,l}^j\left[1+\delta\frac{\sigma^2}{h^2}\right] + v_{i,l}^{j+1}\left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}(v_{i,l}^{j+1}-v_{i,l}^{j-1})\right] + v_{i,l}^{j-1}\left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}(-v_{i,l}^{j+1}+v_{i,l}^{j-1})\right]$$

where: $v_{i,l} = \begin{bmatrix} v(t_i,y_l,S_0) \\ v(t_i,y_l,S_1) \\ \\ \vdots \\ \\ v(t_i,y_l,S_{J-1}) \\ v(t_i,y_l,S_J) \end{bmatrix} \in \mathbb{R}^{J+1}$

The Newton method: For all l in $0,...,L$: $v_{i,l}^{(k+1)} = v_{i,l}^{(k)} - \left[\frac{\partial F}{\partial v_{i,l}^{(k)}}\right]^{-1} F(v_{i,l}^{(k)})$

With the following derivatives

$$\frac{\partial}{\partial v_{i,l}^j}F(v_{i,l}) = 1+\delta\frac{\sigma^2}{h^2}$$
$$\frac{\partial}{\partial v_{i,l}^{j+1}}F(v_{i,l}) = \left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}(2v_{i,l}^{j+1}-v_{i,l}^{j-1})\right]$$
$$\frac{\partial}{\partial v_{i,l}^{j-1}}F(v_{i,l}) = \left[-\frac{\delta\sigma^2}{2h^2}+\psi\frac{\delta\sigma^2}{4h^2}(-v_{i,l}^{j+1}+2v_{i,l}^{j-1})\right]$$

We denote:

$$c_0 = 0$$
$$c_1 = -\frac{\delta\sigma^2}{2h^2} + \psi\frac{\delta\sigma^2}{4h^2}(-v_{i,l}^{j+1} + 2v_{i,l}^{j-1})$$
$$c_2 = 1 + \delta\frac{\sigma^2}{h^2}$$
$$c_3 = -\frac{\delta\sigma^2}{2h^2} + \psi\frac{\delta\sigma^2}{4h^2}(2v_{i,l}^{j-1} - v_{i,l}^{j+1})$$

The Jacobian matrix is: $\frac{\partial F}{\partial v_{i,l}^{(k)}} = \begin{bmatrix} c_0 & 0 & 0 & \cdots & & 0 \\ c_1 & c_2 & c_3 & \ddots & & \vdots \\ 0 & c_1 & c_2 & \ddots & & \\ \vdots & \ddots & \ddots & \ddots & c_3 & 0 \\ & & & c_1 & c_2 & c_3 \\ 0 & \cdots & & 0 & 0 & c_0 \end{bmatrix} \in \mathbb{R}^{J+1,J+1}$

Operator Splitting Idea: At each time of the discretization, add jumps and then solve the linear system.

4 Viscosity Solution

4.1 Existence result

We denote by $C^{1,0,2} := C^{1,0,2}([0,T) \times \mathbb{R} \times \mathbb{R})$ the class of functions $\gamma : [0,T) \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ that are continuously differentiable with respect to the first variable on $[0,T)$, continuous with respect to the second variable on $\mathbb{R}$, twice continuously differentiable with respect to the third variable on $\mathbb{R}$.

Definition 1.

(i) If u is an upper semicontinuous (USC) function on $[0,T) \times \mathbb{R}^2$, we say that u is a viscosity subsolution to the QVI on $[0,T) \times \mathbb{R}^2$ if $\forall(\bar{t},\bar{y},\bar{S}) \in [0,T) \times \mathbb{R}^2$, $\forall\gamma \in C^{1,0,2}$ such that $(u-\gamma)(\bar{t},\bar{y},\bar{S}) = \max_{(t,y,S)\in[0,T)\times\mathbb{R}^2}(u-\gamma)(t,y,S)$, we have:

$$\min\Big\{ -\frac{\partial}{\partial t}\gamma(\bar{t},\bar{y},\bar{S}) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(\bar{t},\bar{y},\bar{S}) \\ - \bar{\lambda}^a(\bar{y},\bar{S})\left[\beta^a(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}-\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right] - \bar{\lambda}^b(\bar{y},\bar{S})\left[\beta^b(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}+\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right], \gamma(\bar{t},\bar{y},\bar{S})\Big\} \le 0$$

(ii) If θ is an lower semicontinuous (LSC) function on $[0,T) \times \mathbb{R}^2$, we say that θ is a viscosity supersolution to the QVI on $[0,T) \times \mathbb{R}^2$ if $\forall(\bar{t},\bar{y},\bar{S}) \in [0,T) \times \mathbb{R}^2$, $\forall\gamma \in C^{1,0,2}$ such that $(\theta-\gamma)(\bar{t},\bar{y},\bar{S}) = \min_{(t,y,S)\in[0,T)\times\mathbb{R}^2}(\theta-\gamma)(t,y,S)$, we have:

$$\min\Big\{ -\frac{\partial}{\partial t}\gamma(\bar{t},\bar{y},\bar{S}) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(\bar{t},\bar{y},\bar{S}) \\ - \bar{\lambda}^a(\bar{y},\bar{S})\left[\beta^a(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}-\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right] - \bar{\lambda}^b(\bar{y},\bar{S})\left[\beta^b(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}+\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right], \gamma(\bar{t},\bar{y},\bar{S})\Big\} \ge 0$$

(iii) If v is locally bounded on $[0,T) \times \mathbb{R}^2$, we say that v is a viscosity solution to QVI on $[0,T) \times \mathbb{R}^2$ if its upper semicontinuous envelope v^* and its lower semicontinuous envelope v_* are respectively subsolution on $[0,T) \times \mathbb{R}^2$ and supersolution on $[0,T) \times \mathbb{R}^2$ to the QVI.

Proposition 1. *The value function v is bounded on $[0,T] \times \mathbb{R}^2$*

Proof.
$\forall(t,y,S) \in [0,T) \times \mathbb{R}^2$, we have

$$v(t,y,S) = \sup_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}\left[\int_t^\tau \beta^a(Y_{u^-},S_u)\,dN_u^a + \int_t^\tau \beta^b(Y_{u^-},S_u)\,dN_u^b\right]$$
$$= \sup_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}\left[\int_t^\tau \beta^a(Y_{u^-},S_u)\,\bar{\lambda}^a(Y_{u^-},S_u)\,\mathbb{1}_{\{Y_{u^-}\ge 2\xi\}} + \beta^b(Y_{s^-},S_u)\,\bar{\lambda}^b(Y_{u^-},S_u)\,du\right]$$

We look at both quantities inside the integrand, we assume that $\mathfrak{r}$ has linear growth and that the process $Y = (Y_t)_{t\in\mathfrak{T}}$ is bounded.

$$\begin{aligned}
\beta^a(Y_{u^-}, S_u)\,\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}} &= \big(\varphi_c(Y_{u^-}-\xi) - \varphi_c(Y_{u^-}) - \xi S_u + \mathfrak{r}(Y_{u^-})\big)\ \max\left\{a_0, a_1 + a_2\,(S_u - \frac{c}{Y_{u^-}^2})\right\}\ \mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\,,\\
&\leq \big(|\varphi_c(Y_{u^-}-\xi)| + |\varphi_c(Y_{u^-})| + |\xi S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(|a_0| + |a_1 + a_2\,(S_u - \frac{c}{Y_{u^-}^2})|\right)\\
&\leq \big(|\frac{c}{Y_{u^-}-\xi}| + |\frac{c}{Y_{u^-}}| + |\xi S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(|a_0| + |a_1 + |a_2|\,(|S_u| + \frac{c}{Y_{u^-}^2})\right)\\
&\leq \big(\frac{c}{\xi} + \frac{c}{2\xi} + \xi|S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(a_0 + a_1 + a_2\,(|S_u| + \frac{c}{\xi^2})\right)\\
&\leq C_0^a + C_1^a|Y_{u^-}| + C_2^a \sup_{u\in[t,\tau]}|S_u| + C_3^a \sup_{u\in[t,\tau]}|S_u|^2
\end{aligned}$$

Where C_0^a, C_1^a, C_2^a, C_3^a are constant depending on ξ, c, a_0, a_1 and a_2

$$\begin{aligned}
\beta^b(Y_{u^-}, S_u)\,\bar{\lambda}^b(Y_{u^-}, S_u) &= \big(\varphi_c(Y_{u^-}+\xi) - \varphi_c(Y_{u^-}) + \xi S_u + \mathfrak{r}(Y_{u^-})\big)\ \max\left\{a_0, a_1 + a_2\,(\frac{c}{Y_{u^-}^2} - S_u)\right\}\\
&\leq \big(|\varphi_c(Y_{u^-}+\xi)| + |\varphi_c(Y_{u^-})| + |\xi S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(|a_0| + |a_1 + a_2\,(\frac{c}{Y_{u^-}^2} - S_u|\right)\\
&\leq \big(|\frac{c}{Y_{u^-}+\xi}| + |\frac{c}{Y_{u^-}}| + |\xi S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(|a_0| + |a_1 + a_2\,(\frac{c}{Y_{u^-}^2} - S_u)|\right)\\
&\leq \big(|\frac{c}{Y_{u^-}+\xi}| + |\frac{c}{Y_{u^-}}| + \xi|S_u| + |\mathfrak{r}(Y_{u^-})|\big)\ \left(a_0 + a_1 + a_2\,(\frac{c}{Y_{u^-}^2} + |S_u|)\right)\\
&\leq C_0^b + C_1^b|Y_{u^-}| + C_2^b \sup_{u\in[t,\tau]}|S_u| + C_3^b \sup_{u\in[t,\tau]}|S_u|^2
\end{aligned}$$

Using Jensen for concave functions and then Doob inequality we can show that

$$\begin{aligned}
\mathbb{E}\left[\sup_{u\in[t,T]}|S_u|\right] &\leq \mathbb{E}\left[\sup_{u\in[0,T]}|S_u|\right] \leq |S_0| + \sigma\mathbb{E}\left[\sup_{u\in[0,T]}|\int_0^u dW_s|\right] = |S_0| + \sigma\mathbb{E}\left[\sup_{u\in[0,T]}|W_u|\right] = |S_0| + \sigma\mathbb{E}\left[\sup_{u\in[0,T]}\sqrt{W_u^2}\right]\\
&\leq |S_0| + 2\sigma\sqrt{T}\\
\mathbb{E}\left[\sup_{u\in[t,T]}|S_u|^2\right] &\leq \mathbb{E}\left[\sup_{u\in[0,T]}|S_u|^2\right] = S_0^2 + 2S_0\sigma\mathbb{E}\left[\sup_{u\in[0,T]}W_u\right] + \sigma^2\mathbb{E}\left[\sup_{u\in[0,T]}W_u^2\right]\\
&\leq S_0^2 + 4S_0\sigma^2\sqrt{T} + 4\sigma^2 T
\end{aligned}$$

We can conclude that the value function has linear growth and then is locally bounded from above. Furthermore, looking at the QVI the value function is bounded from below by 0.

Proposition 2. *v is a viscosity subsolution to the QVI on $[0,T)\times\mathbb{R}^2$.*

Proof.

v is bounded on $[0,T]\times\mathbb{R}^2$ so we can define v^* its upper semicontinuous envelope. Let $(\bar{t},\bar{y},\bar{S}) \in [0,T)\times\mathbb{R}^2$ and $\gamma \in C^{1,0,2}$ such that:

$$0 = (v^* - \gamma)(\bar{t},\bar{y},\bar{S}) = \max_{(t,y,S)\in[0,T)\times\mathbb{R}^2}(v^* - \gamma)(t,y,S)$$

We assume this maximum to be strict. By definition of $v^*(\bar{t},\bar{y},\bar{S})$, their exists $(t_m, y_m, S_m)_m$ a sequence of $[0,T)\times\mathbb{R}^2$ such that:

$$\begin{aligned}
(t_m, y_m, S_m) &\xrightarrow{m\to\infty} (\bar{t},\bar{y},\bar{S})\\
v(t_m, y_m, S_m) &\xrightarrow{m\to\infty} v^*(\bar{t},\bar{y},\bar{S})
\end{aligned}$$

We prove the result by contradiction. Assume there exists $\eta > 0$ such that

$$\min\Big\{ -\frac{\partial}{\partial t}\gamma(\bar{t},\bar{y},\bar{S}) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(\bar{t},\bar{y},\bar{S}) \\ - \bar{\lambda}^a(\bar{y},\bar{S})\left[\beta^a(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}-\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right] - \bar{\lambda}^b(\bar{y},\bar{S})\left[\beta^b(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}+\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right], \gamma(\bar{t},\bar{y},\bar{S})\Big\} > \eta$$

This means that both quantities in the maximum must satisfy

$$\begin{cases} -\frac{\partial}{\partial t}\gamma(\bar{t},\bar{y},\bar{S}) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(\bar{t},\bar{y},\bar{S}) - \bar{\lambda}^a(\bar{y},\bar{S})\left[\beta^a(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}-\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right] - \bar{\lambda}^b(\bar{y},\bar{S})\left[\beta^b(\bar{y},\bar{S}) + \gamma(\bar{t},\bar{y}+\xi,\bar{S}) - \gamma(\bar{t},\bar{y},\bar{S})\right] > \eta \\ \gamma(\bar{t},\bar{y},\bar{S}) > \eta \end{cases} \tag{1}$$

Then, as γ is $C^{1,0,2}$, we must have

$$\min\Big\{ -\frac{\partial}{\partial t}\gamma(t,y,S) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(t,y,S) \\ - \bar{\lambda}^a(y,S)\left[\beta^a(y,S) + \gamma(t,y-\xi,S) - \gamma(t,y,S)\right] - \bar{\lambda}^b(y,S)\left[\beta^b(y,S) + \gamma(t,y+\xi,S) - \gamma(t,y,S)\right], \gamma(t,y,S)\Big\} > \eta$$

on $B := B\left((\bar{t},\bar{y},\bar{S}),\varepsilon\right)$, where $\varepsilon := \min_{(t,y,S)\in A} d\left((t,y,S),(\bar{t},\bar{y},\bar{S})\right)$ and $A := \left\{(t,y,S)\in[0,T)\times\mathbb{R}^2 \;\middle|\; |v^*(t,y,S) - \gamma(t,y,S)| \geq \eta\right\}$. Without loss of generality, we can assume that B contains the sequence $(t_m, y_m, S_m)_m$

We introduce a stopping time $\pi_m = \inf\left\{t \geq t_m \mid (t, Y_t^{t_m,y_m}, S_t^{t_m,S_m}) \notin B\right\}$

By Itô's formula applied to γ at time $\pi_m \wedge \tau$ where τ is a stopping time,

$$\begin{aligned} \gamma(\pi_m\wedge\tau, Y_{\pi_m\wedge\tau}^{t_m,y_m}, S_{\pi_m\wedge\tau}^{t_m,S_m}) = \gamma(t_m,y_m,S_m) &+ \int_{t_m}^{\pi_m\wedge\tau}\frac{\partial}{\partial t}\gamma(u,Y_{u^-},S_u)\,du + \frac{1}{2}\sigma^2\int_{t_m}^{\pi_m\wedge\tau}\frac{\partial^2}{\partial S^2}\gamma(u,Y_{u^-},S_u)\,du \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\frac{\partial}{\partial S}\gamma(u,Y_{u^-},S_u)\,dW_u \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u,Y_{u^-}-\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right)\,dN_u^a \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u,Y_{u^-}+\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right)\,dN_u^b \end{aligned}$$

$$\begin{aligned} \gamma(\pi_m\wedge\tau, Y_{\pi_m\wedge\tau}^{t_m,y_m}, S_{\pi_m\wedge\tau}^{t_m,S_m}) = \gamma(t_m,y_m,S_m) &+ \int_{t_m}^{\pi_m\wedge\tau}\Big[\frac{\partial}{\partial t}\gamma(u,Y_{u^-},S_u) + \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(u,Y_{u^-},S_u) \\ &\quad + \bar{\lambda}^a(Y_{u^-},S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\left(\gamma(u,Y_{u^-}-\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right) \\ &\quad + \bar{\lambda}^b(Y_{u^-},S_u)\left[\gamma(u,Y_{u^-}+\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right]\Big]\,du \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u,Y_{u^-}-\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right]\,d\tilde{N}_u^a \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u,Y_{u^-}+\xi,S_u) - \gamma(u,Y_{u^-},S_u)\right]\,d\tilde{N}_u^b \\ &+ \int_{t_m}^{\pi_m\wedge\tau}\frac{\partial}{\partial S}\gamma(u,Y_{u^-},S_u)\,dW_u \end{aligned}$$

$$\gamma(\pi_m \wedge \tau, Y^{t_m,y_m}_{\pi_m\wedge\tau}, S^{t_m,S_m}_{\pi_m\wedge\tau}) = \gamma(t_m, y_m, S_m) - \int_{t_m}^{\pi_m\wedge\tau} -\Big\{\frac{\partial}{\partial t}\gamma(u, Y_{u^-}, S_u) + \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(u, Y_{u^-}, S_u)$$
$$+ \bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\left[\beta^a(Y_{u^-}, S_u) + \gamma(u, Y_{u^-} - \xi, S_u) - \gamma(u, Y_{u^-}, S_u)\right]$$
$$+ \bar{\lambda}^b(Y_{u^-}, S_u)\left[\beta^b(Y_{u^-}, S_u) + \gamma(u, Y_{u^-} + \xi, S_u) - \gamma(u, Y_{u^-}, S_u)\right]\Big\}du$$
$$- \int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du$$
$$+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u, Y_{u^-} - \xi, S_u) - \gamma(u, Y_{u^-}, S_u)\right]d\tilde{N}^a_u$$
$$+ \int_{t_m}^{\pi_m\wedge\tau}\left[\gamma(u, Y_{u^-} + \xi, S_u) - \gamma(u, Y_{u^-}, S_u)\right]d\tilde{N}^b_u$$
$$+ \int_{t_m}^{\pi_m\wedge\tau}\frac{\partial}{\partial S}\gamma(u, Y_{u^-}, S_u)\,dW_u$$

Inside the first integrand, we recognize the stochastic partial differential equation that we supposed greater than $\eta > 0$ and the last three terms are martingale under the probability measure $\mathbb{P}$. Therefore, taking the expectation of the above quantity leads to

$$\mathbb{E}\left[\gamma(\pi_m \wedge \tau, Y^{t_m,y_m}_{\pi_m\wedge\tau}, S^{t_m,S_m}_{\pi_m\wedge\tau})\right] \leq \gamma(t_m, y_m, S_m) - \mathbb{E}\left[\int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du\right]$$

That we can write also

$$\gamma(t_m, y_m, S_m) \geq \mathbb{E}\left[\gamma(\pi_m \wedge \tau, Y^{t_m,y_m}_{\pi_m\wedge\tau}, S^{t_m,S_m}_{\pi_m\wedge\tau}) + \int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du\right]$$

As γ is $C^{1,0,2}$, then $\gamma(t_m, y_m, S_m) \xrightarrow{m\to\infty} \gamma(\bar{t}, \bar{y}, \bar{S}) = v^*(\bar{t}, \bar{y}, \bar{S})$. We also have $v(t_m, y_m, S_m) \xrightarrow{m\to\infty} v^*(\bar{t}, \bar{y}, \bar{S})$. Therefore, since B is compact we can find a m sufficiently large such as $v(t_m, y_m, S_m) + \frac{\eta}{2} > \gamma_m$

$$v(t_m, y_m, S_m) \geq -\frac{\eta}{2} + \mathbb{E}\left[\gamma(\pi_m \wedge \tau, Y^{t_m,y_m}_{\pi_m\wedge\tau}, S^{t_m,S_m}_{\pi_m\wedge\tau}) + \int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du\right]$$

Moreover,

$$\gamma(\pi_m \wedge \tau, Y^{t_m,y_m}_{\pi_m\wedge\tau}, S^{t_m,S_m}_{\pi_m\wedge\tau}) = \underbrace{\gamma(\pi_m, Y^{t_m,y_m}_{\pi_m}, S^{t_m,S_m}_{\pi_m})}_{\geq v(\pi_m, Y^{t_m,y_m}_{\pi_m}, S^{t_m,y_m}_{\pi_m})+\eta}\mathbb{1}_{\{\pi_m<\tau\}} + \underbrace{\gamma(\tau, Y^{t_m,y_m}_{\tau}, S^{t_m,S_m}_{\tau})}_{>\eta}\mathbb{1}_{\{\pi_m\geq\tau\}}$$

Putting all things together

$$v(t_m, y_m, S_m) \geq \frac{\eta}{2} + \mathbb{E}\left[\int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du + v(\pi_m, Y^{t_m,y_m}_{\pi_m}, S^{t_m,y_m}_{\pi_m})\mathbb{1}_{\{\pi_m<\tau\}}\right]$$

By taking the supremum over all the stopping time in $\mathcal{T}_{t_m,T}$ on the right-hand side, we get

$$v(t_m, y_m, S_m) > \sup_{\tau\in\mathcal{T}_{t_m,T}}\mathbb{E}\left[\int_{t_m}^{\pi_m\wedge\tau}\left[\beta^b(Y_{u^-}, S_u)\bar{\lambda}^b(Y_{u^-}, S_u) + \beta^a(Y_{u^-}, S_u)\bar{\lambda}^a(Y_{u^-}, S_u)\,\mathbb{1}_{\{Y_{u^-}\geq 2\xi\}}\right]du + v(\pi_m, Y^{t_m,y_m}_{\pi_m}, S^{t_m,y_m}_{\pi_m})\mathbb{1}_{\{\pi_m<\tau\}}\right]$$

This contradicts the dynamic programming principle. Necessarily, we deduce:

$$\min\Big\{-\frac{\partial}{\partial t}\gamma(\bar{t}, \bar{y}, \bar{S}) - \frac{1}{2}\sigma^2\frac{\partial^2}{\partial S^2}\gamma(\bar{t}, \bar{y}, \bar{S})$$
$$- \bar{\lambda}^a(\bar{y}, \bar{S})\left[\beta^a(\bar{y}, \bar{S}) + \gamma(\bar{t}, \bar{y} - \xi, \bar{S}) - \gamma(\bar{t}, \bar{y}, \bar{S})\right] - \bar{\lambda}^b(\bar{y}, \bar{S})\left[\beta^b(\bar{y}, \bar{S}) + \gamma(\bar{t}, \bar{y} + \xi, \bar{S}) - \gamma(\bar{t}, \bar{y}, \bar{S})\right], \gamma(\bar{t}, \bar{y}, \bar{S})\Big\} \leq 0$$

and v is a viscosity subsolution to the QVI on $[0, T)x\mathbb{R}^2$

Definition 2. *The value function v is said to have at most quadratic growth in its third variable if there exists a constant $C > 0$ such*

that $\forall S \in \mathbb{R}$, $v(t, y, S) \leq C(1 + S^2)$ *for any fixed but arbitrary* $t \in [0, T]$ *and* $y \in Q$.

We assume without loss of generality that Y_0, $\underline{Y}$, $\overline{Y}$ are multiple of ξ.